

Block-Varying Imperfect CSI and Residual SIC for Active STAR-RIS Assisted UAV Networks With NOMA

Sheng Hao, *Member, IEEE*, Yinyi Chen, Xiying Fan, Hao Xu, *Senior Member, IEEE*, Dianxin Luan, Ji Wang, *Senior Member, IEEE*, Cong Li and Zongpeng Li, *Senior Member, IEEE*

I. FIXED-PARAMETER MODELING OF IMPERFECT CSI AND RESIDUAL SIC

In practical UAV communication environments, the imperfect CSI and residual SIC interference may vary over time and environmental conditions. In the main paper, the imperfect CSI parameter σ_e^2 and the residual SIC factor β_{RI} are modeled as fixed parameters. The reason is that, following the existing literature, such as **Ref.[1]**, **Ref.[2]** and [37], imperfect CSI and residual SIC interference are commonly modeled as fixed parameters in order to preserve the tractability of analytical derivations. This treatment has been widely adopted in the analytical modeling of STAR-RIS assisted and NOMA-assisted communication systems. On the other hand, the introduction of the CSI error variance and the residual SIC factor itself already indicates that the channel estimation error and SIC residue are treated as random and non-ideal effects in a statistical sense, rather than as deterministic or perfect terms. Hence, the adopted fixed-parameter treatment is not meant to neglect their randomness, but to capture their average influence in a mathematically tractable manner, while still allowing us to clearly investigate the impact of imperfect CSI and residual SIC on the downlink performance.

II. BLOCK-VARYING SETTING

To further discuss the dynamic characteristics of imperfect CSI and residual SIC interference, we consider a block-varying setting. Specifically, for each transmission block ℓ , the channel estimation error variance σ_e^2 and the residual interference β_{RI} associated with SIC process are randomly generated as

$$\sigma_e^2(\ell) \sim \mathcal{U}[\sigma_{e,\min}^2, \sigma_{e,\max}^2], \quad \beta_{RI}(\ell) \sim \mathcal{U}[\beta_{RI,\min}, \beta_{RI,\max}] \quad (1)$$

Then, $\sigma_e^2(\ell)$ and $\beta_{RI}(\ell)$ are substituted into the corresponding SINR expressions of our work. The final performance metric is obtained by averaging over all transmission blocks. Taking CP_κ as an example, it can be computed as

$$\bar{CP} * \kappa = \frac{1}{L} \sum * \ell = 1^L CP_\kappa(\sigma_e^2(\ell), \beta_{RI}(\ell)) \quad (2)$$

where L denotes the number of transmission blocks. Under this setting, the block-level dynamic characteristics of imperfect CSI and residual SIC interference can be approximately captured without changing the main derivation method.

III. PERFORMANCE EVALUATION

Using the same parameters of the second experiment group (Section III.B), while additionally setting $\sigma_{e,\min}^2 = 0.001$,

$\sigma_{e,\max}^2 = 0.01$, $\beta_{RI,\min} = 0.01$, $\beta_{RI,\max} = 0.1$ and $L = 20$, **fig. 1 illustrates the impact of number of active STAR-RIS components N on the downlink performance of active STAR-RIS assisted UAV networks with NOMA under ES protocol, where imperfect CSI and residual SIC interference vary across transmission blocks.**

It is easy to find that the main performance trends under dynamic $\sigma_e^2(\ell)$ and $\beta_{RI}(\ell)$ remain consistent with those under the fixed-parameter case. Specifically, as the number of active STAR-RIS components N increases, the coverage probability CP , ergodic capacity EC and energy efficiency EE increase, while the delay outage rate DOR decreases. This indicates that introducing block-level random variation of imperfect CSI and residual SIC interference does not change the essential conclusion of our paper. Moreover, compared with the fixed-parameter case, the CP , EC and EE under the block-random case are lower, whereas the DOR becomes higher. The reason is that the random fluctuation of $\sigma_e^2(\ell)$ and $\beta_{RI}(\ell)$ introduces additional uncertainty into the transmission process, which causes performance loss after block averaging.

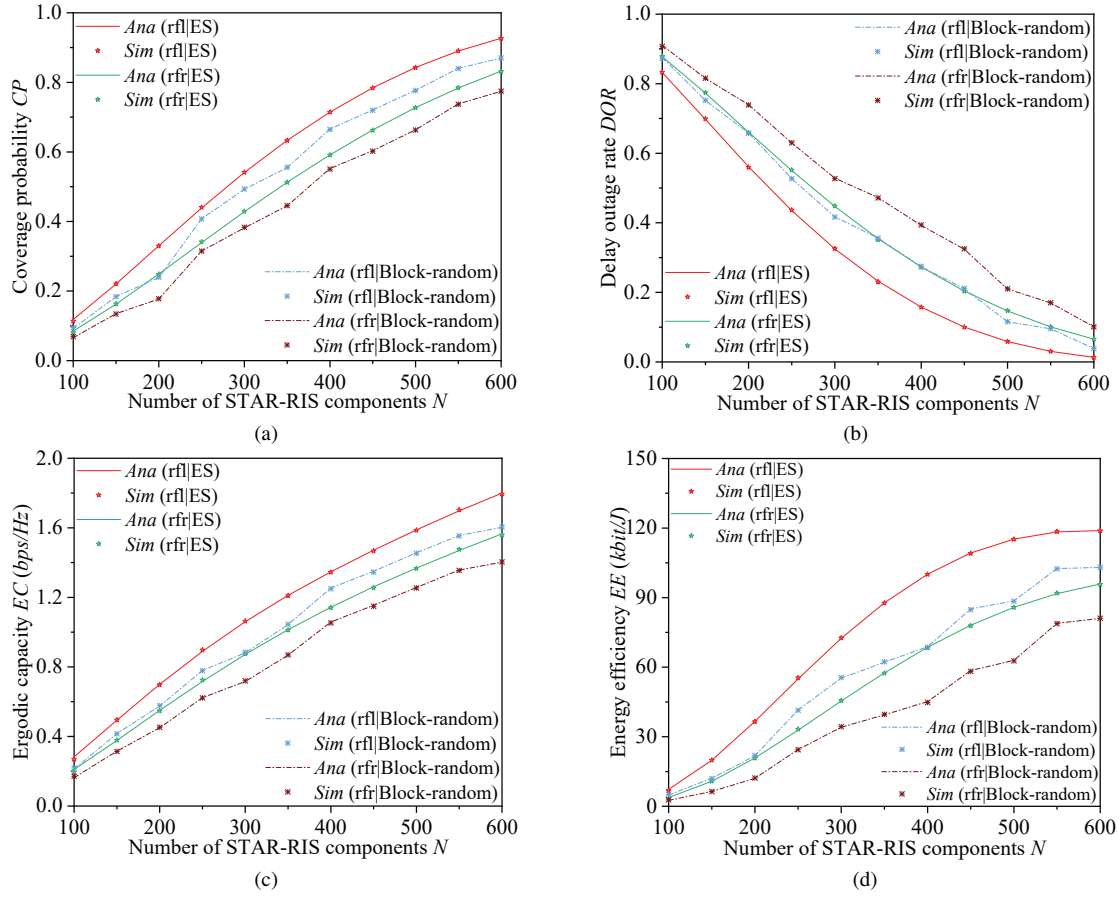
It is also worth noting that the curves under the block-random setting exhibit mild fluctuations. This is because the performance metrics are averaged over a finite number of transmission blocks with randomly generated $\sigma_e^2(\ell)$ and $\beta_{RI}(\ell)$. Hence, there exist slight variations compared with the deterministic case. Nevertheless, the overall performance trends remain unchanged.

Through comparing the simulation and numerical analysis results, it can be drawn that our proposed model is still accurate for active STAR-RIS assisted UAV networks with NOMA under the considered block-varying setting. This further demonstrates that the proposed analytical framework can reflect the average impact of time-varying and environment-dependent imperfect CSI and residual SIC characteristics.

Ref.[1] and **Ref.[2]** are listed as follows:

Ref.[1]: S. Bisen, P. Shaik and V. Bhatia, "On Performance of Energy Harvested Cooperative NOMA Under Imperfect CSI and Imperfect SIC," *IEEE Trans. Veh. Technol.*, vol. 70, no. 9, pp. 8993-9005, Sep. 2021.

Ref.[2]: A. Baghel, A. Singh Parihar, V. Bhatia, O. Krejcar and K. Choi, "On Performance of NOMA-Empowered HCN: A Stochastic Geometry Approach Considering Imperfect SIC and CSI," *IEEE Wireless Commun. Lett.*, vol. 14, no. 7, pp. 1924-1928, Jul. 2025.

Fig. 1. Our work versus varying σ_e^2 and β_{RI}